Barret Schloerke

Senior Software Engineer, Posit
`code[c("architect", "manager", "philosopher")]`
`data[c("wrangler", "explorer", "visualizer")]`

schloerke@gmail.com
schloerke.com
schloerke
@schloerke.com
0000-0001-9986-114X
schloerke

Summary

Senior software engineer for Shiny, Posit's web framework for R and Python (1M+ monthly downloads). My statistician's skepticism (PhD) shapes how I build and test tools that turn data scientists into full-stack app developers. Highlights: built OpenTelemetry tracing into Shiny for R and Python, so users can see where their apps spend time; created shinyreact, so Shiny apps can use React-based UI; and cut Shiny's release testing from 3 weeks for 10 people to under 2 days for 2.

Experience

Senior Software Engineer, Posit · Remote — Jan 2022 – Present

- Core developer of Shiny for R and Python, Posit's flagship framework that brings users to its commercial products.
- Cut CI time 98%, measured by 4+ hours of testing finishing in a 4.5-minute run, by architecting Playwright controllers for every Shiny for Python component and spreading 1000+ tests over 170 jobs (5 Pythons × 3 browsers).
- Brought Shiny for R's best UI to Python by porting bslib's components: sidebars, cards, value boxes, tooltips, and more.
- Built shinylive for R and Python: Shiny apps that run in the browser with no server, even in Quarto sites and slides.
- Enabled every Shiny user to trace apps with OpenTelemetry, build LLM-powered apps (querychat, shinychat), and pair agentic coding with React-based UI (shinyreact).

Software Engineer, RStudio (now Posit) · Remote — Jan 2018 – Dec 2021

- Cut release testing from 3 weeks across 10 people to under 2 days for 2 by architecting shinycoreci: nightly testing of 150+ apps on Mac, Windows, and Linux × 5 R versions. Built shiny-workflows, shared CI for every Shiny team package.
- Migrated Shiny's JavaScript to TypeScript, so every PR's client code is type-checked across all files (before, none was).
- Sustained plumber, reactlog, shinytest2, and dozens of other Shiny packages by owning releases, docs, and triage.

Tool Development Team, Gates Foundation: HBGDki · Remote — Apr 2016 – Dec 2018

- Produced custom summary statistics for 75+ datasets, feeding visualization apps that consultants used in work to prevent avoidable infant deaths.
- Protected sensitive study data by coordinating secure data-handling practices with five other data scientists.
- Standardized the team's R package development by creating and maintaining template R packages.

Researcher, DARPA XDATA and D3M grants, Purdue University · West Lafayette, IN — Sep 2016 – Dec 2017

- Delivered a web interface for subject matter experts to explore machine learning pipelines by partnering remotely with Hafen Consulting, Kitware, and KnowledgeVis.
- Led service integration across 15 partner working groups by serving as the lead point of contact.

Software Engineer, Metamarkets · San Francisco, CA — Jan 2011 – Jul 2012

- Built interactive D3 visualizations for analytics dashboards, backed by Druid, on DVL, a reactive data visualization library, in CoffeeScript.
- Wrote Node.js front-end servers that automated deploys, clusters, and company / user management, allowing anyone on the team to ship or debug the client experience.
- Reduced site startup time and server load by trimming data queries and startup work.

Data Scientist Intern, Novartis Pharmaceutical R&D · Basel, Switzerland — May 2008 – Aug 2008

- Equipped management with a live view of employee availability by building a Gantt chart tool.
- Standardized common graphics for the R&D team by writing a web tutorial that packaged them as reusable plotting layers.

Education

PhD, Statistics, Purdue University · West Lafayette, IN — 2013 – 2017
- Advisors: William Cleveland and Ryan Hafen. Emphasis: large data visualization.
- National Science Foundation Graduate Research Fellow (2012 – 2017).

MS, Mathematical Statistics, Purdue University · West Lafayette, IN — 2013 – 2014

PhD program, Statistics, Rice University · Houston, TX — 2012
- Advisor: Hadley Wickham.

BS, Computer Engineering, Iowa State University · Ames, IA — 2006 – 2010
- Statistics graphics research with Dianne Cook and Heike Hofmann.

Open source software

2,300+ merged pull requests across 170+ GitHub repositories, and 1,400+ reviews of others' pull requests, since 2018. Packages I maintain or author, by merged PRs:

`shiny` Python · author · 370+ PRs · A web development framework for Python.
- Agent Skills (2026): Skills shipped in the package so coding agents know how to write Shiny apps.
- Test mode (2026): test_server() and exported test values for fast tests without a browser.
- Session lifecycle (2026): session.destroy(), a soft teardown when a session closes, and session.allow_reconnect().
- Release train skill (2026): An Agent Skill that walks a py-shiny release from prep through PyPI, conda-forge, and shinylive.
- OpenTelemetry (2026): Traces and logs for sessions, reactivity, and outputs.
- Bookmarking (2025): Save and restore an app's state in the URL or on the server.
- Data frames (2024): render.data_frame with editing, selection, sorting, filtering, and styles.
- Output renderers (2024): shiny.render.renderer, a base class for writing your own output renderers.
- API reference (2023 – 2026): The quartodoc function reference and its runnable examples.
- UI components from bslib (2023 – 2024): Ported sidebars, cards, accordions, value boxes, tooltips, and popovers from R's bslib.
- Playwright testing (2023 – 2024): shiny.playwright controllers for every UI component, and the shiny.pytest plugin.

`shinyreact` R, Python, TypeScript · maintainer · 170+ PRs · Client-Side 'React' Interface for 'shiny'
- TypeScript: npm package (2026): A dual build published as @posit-dev/shinyreact, usable with no server-side JS.
- TypeScript: Wire protocol (2026): One versioned protocol, checked by a handshake, shared with the R and Python packages.
- Python: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Agent Skills (2026): Skills for building and converting apps with coding agents, shipped in the package.
- R: Initial development (2026): Ported the Python API to R, then CRAN checks, a pkgdown site, and webR builds for 0.1.0.
- Python: Renderer dependencies (2026): Find and serve the HTML dependencies of Shiny outputs used inside React.
- Python: ui.tsx apps (2026): Write the whole page in React and serve it from Python.
- R: ui.tsx apps (2026): Write the whole page in React and serve it from R.
- TypeScript: React hooks (2026): useShinyInput / useShinyOutput hooks, and ShinyOutput for classic Shiny outputs.
- Python: Testing (2026): Playwright e2e tests, WireTap for wire payloads, and in-memory test_server().
- Python: Initial development (2026): Began as shinyjson, a JSON spec renderer, then turned React-first and dropped json-render.

`GGally` R · maintainer · 150+ PRs · Extension to 'ggplot2'
- S7 (2025): Moved ggmatrix to S7 for ggplot2 4.0.
- Discrete stats (2020): Count, cross, table, and statistic plots for GGally 2.0.
- ggduo, ggnostic, and friends (2016 – 2017): Plot matrices for two groups of variables, model diagnostics, and time series.
- ggpairs and ggmatrix (2014 – 2017): The generalized pairs plot and the plot matrix behind it.

`shinytest2` R · maintainer · 140+ PRs · Testing for Shiny Applications
- Migration from shinytest (2022): migrate_from_shinytest() converts existing shinytest tests.
- Screenshots (2021 – 2023): Consistent screenshots, compared with a kernel density smoother to ignore stray pixels.
- Test recorder (2021 – 2022): record_test() writes a test while you click around the app.
- AppDriver (2021 – 2022): Drive a Shiny app in headless Chrome: set inputs, click, and snapshot values.

`shiny` R, TypeScript · author · 140+ PRs · Web Application Framework for R
- R: MCP Apps (2026): Serve Shiny apps as MCP Apps, with author tools for LLM clients.
- R: Session lifecycle (2026): session$destroy(), a soft teardown when a session closes, and clean state on reconnect.
- R: OpenTelemetry (2025): Traces and logs for sessions, reactivity, and outputs through {otel}.
- R: Quosures (2021): exprToFunction() and installExprFunction() accept quosures, including inject()ed ones.
- TypeScript: TypeScript (2021 – 2022): Ported shiny.js to TypeScript, moved the build to esbuild, and published the types.
- R: devmode() (2020 – 2021): One switch that turns on the options you want while developing an app.
- R: testServer() (2020): Test server functions and modules in memory against a mock session.
- R: Plot brushing (2018): Plot click and brush events carry CSS and image coordinates, and brushing works in WebKit.
- R: reactlog integration (2018 – 2020): Record every reactive event so reactlog can draw the reactive graph.

`plumber` R · maintainer · 130+ PRs · An API Generator for R
- pr_*() API (2020): The pipe-friendly router API and consistent R6 methods of plumber 1.0.
- Docker (2020 – 2021): Docker images for running plumber APIs.
- Async (2019 – 2020): Promise and future support for long-running endpoints.
- Serializers and parsers (2019 – 2020): Body parsers chosen by content type, and serializers for many output formats.
- OpenAPI docs (2018 – 2020): OpenAPI v3 specs and pluggable docs pages (Swagger, RapiDoc, Redoc).

`learnr` R · author · 70+ PRs · Interactive Tutorials for R
- Exercises (2019 – 2020): Safer exercise evaluation, checker functions, and clearer errors.
- Quiz questions (2019 – 2020): Questions rebuilt as Shiny modules, with a custom question API.

`promises` R · maintainer · 40+ PRs · Abstractions for Promise-Based Asynchronous Programming

`leaflet` R · author · 40+ PRs · Create Interactive Web Maps with the JavaScript 'Leaflet' Library

`reactlog` R · maintainer · 40+ PRs · Reactivity Visualizer for 'shiny'

`shinyloadtest` R · maintainer · 30+ PRs · Load Test Shiny Applications

`htmltools` R · author · 30+ PRs · Tools for HTML

`shinylive` R · maintainer · 20+ PRs · Run 'shiny' Applications in the Browser

`shinyswatch` Python · author · 20+ PRs · Bootswatch + Bootstrap 5 themes for Shiny.

`chromote` R · author · 17 PRs · Headless Chrome Web Browser Interface

`sortable` R · author · 12 PRs · Drag-and-Drop in 'shiny' Apps with 'SortableJS'

`trelliscopejs` R · author · 11 PRs · Create Interactive Trelliscope Displays

`lmmpar` R · author · 9 PRs · Parallel Linear Mixed Model

`swagger` R · author · 9 PRs · Dynamically Generates Documentation from a 'Swagger' Compliant API

`leaflet.providers` R · maintainer · 8 PRs · Leaflet Providers

`autocogs` R · author · 8 PRs · Automatic Cognostic Summaries

`shinychat` R, Python · author · 8 PRs · Chat UI Component for 'shiny'

`lazyrmd` R · author · 7 PRs · Render R Markdown Outputs Lazily

`leaflet.extras` R · author · 7 PRs · Extra Functionality for 'leaflet' Package

`gqlr` R · maintainer · 7 PRs · 'GraphQL' Server in R

`packagedocs` R · author · 6 PRs · Build Website of Package Documentation

`sass` R · author · 6 PRs · Syntactically Awesome Style Sheets ('Sass')

`querychat` R, Python · author · 5 PRs · Filter and Query Data Frames in 'shiny' Using an LLM Chat Interface

`leaflet.esri` R · author · 4 PRs · 'ESRI' Bindings for the 'leaflet' Package

`gt` R · author · 3 PRs · Easily Create Presentation-Ready Display Tables

Also contributed to: bslib, shinylive, htmltools, posit-sdk, shinytest, httpuv, meetupr, rmarkdown, chatlas, websocket, flexdashboard, webshot2, cachem, later, vetiver, testthat, trelliscope, webshot, webdriver, fastmap, narwhals, plumber2, graphql, ggplot2, redoc, shinyvalidate, jquerylib, routr, brand.yml.

Earlier research software

`rbokeh`, Purdue University · R interface to the Bokeh visualization library. 2016

`DVL`, Metamarkets · Functionally reactive JavaScript library binding data to visualizations. 2011 – 2012

`helpr`, Rice University · Friendlier HTML documentation for R, with search across packages. 2010

`cranvas`, Iowa State University · Interactive plotting canvas for R, drawing a million points in under a second. 2010

`tourr`, Iowa State University · Geodesic interpolation and basis generation for tour methods in R. 2009

`DescribeDisplay`, Iowa State University · Publication-quality R graphics from GGobi plots. 2009

`geozoo`, Iowa State University · Library of high-dimensional geometric objects, including the new Hyper Ring Torus. 2007

Publications

Gökalp Yavuz, F., and **B. Schloerke** (2020). Parallel computing in linear mixed models. *Computational Statistics 35(3): 1273–1289*.

Schloerke, B., H. Wickham, D. Cook, and H. Hofmann (2016). Escape from Boxland: Generating a Library of High-Dimensional Geometric Shapes. *The R Journal 8(2): 243–257*.

Emerson, J., W. Green, **B. Schloerke**, J. Crowley, D. Cook, H. Hofmann, and H. Wickham (2013). The Generalized Pairs Plot. *Journal of Computational and Graphical Statistics 22(1)*.

Hofmann, H., D. Cook, C. Kielion, **B. Schloerke**, J. Hobbs, A. Loy, L. Mosley, D. Rockoff, Y. Huang, D. Wrolstad, and T. Yin (2011). Delayed, Canceled, on Time, Boarding... Flying in the USA. *Journal of Computational and Graphical Statistics 20(2)*.

Mosley, L., D. Cook, H. Hofmann, C. Kielion, and **B. Schloerke** (2010). Monitoring the 2008 Election Visually. *Chance 23(3)*.

Talks & workshops

Building Custom Shiny UI with React · Johnson & Johnson Shiny Day (remote) Oct 2026

Beyond Bootstrap: Building Custom Shiny UI with React · posit::conf(2026) Sep 2026

OpenTelemetry in R: Capturing Traces, Logs, and Metrics for Shiny Applications · Open Source in Pharma Apr 2026

{GGally}: Success and Long-Haul Maintenance Strategies · ggplot2 extenders · video Nov 2025

Observability at scale: Monitoring Shiny Applications with OpenTelemetry · posit::conf(2025) · video Sep 2025

{plumber2}: Streamlining Web API Development in R · useR! 2025 · video Aug 2025

{querychat}: Natural language interfaces for interactive data graphics in Shiny · JSM 2025 Aug 2025

AI Kung-Fu : Training your LLM client in two function calls · ShinyConf 2025 Apr 2025

{shinylive}: Serverless Shiny applications workshop. An exercise in deploying your app to GitHub Pages · R/Pharma 2024 · video Oct 2024

Data frames in Shiny for Python · posit::conf(2024) · video Aug 2024

{shinylive}: Serverless Shiny apps · ShinyConf 2024 · video Apr 2024

Lessons learned from testing 2500+ Shiny Apps every day · ShinyConf 2023 · video Mar 2023

{shinytest2}: Regression testing Shiny applications · Johnson & Johnson Shiny Day Oct 2022

{shinytest2}: Regression testing Shiny applications · rstudio::conf(2022) · video Jul 2022

{shinytest2}: Regression Testing for Shiny Applications · R/Pharma Shiny for FDA Submissions · video May 2022

shinytest2: Regression Testing for Shiny Applications · ShinyConf 2022 · video Apr 2022

plumber + future: Async Web APIs · Harvard R User Group · video Aug 2021

An intro to plumber · Harvard R User Group · video Aug 2021

plumber + future: Async Web APIs · rstudio::global(2021) · video Jan 2021

Expanding R Horizons: Integrating R with Plumber APIs · RStudio Webinar · video Oct 2020

Episode 14: Shining a Light on learnr · Shiny Developer Series · video Oct 2020

Episode 13: Inside Plumber 1.0 · Shiny Developer Series · video Sep 2020

Episode 12: reactlog · Shiny Developer Series · video Sep 2020

Reactlog 2.0: Debugging the State of Shiny · Statistical Programming DC Apr 2020

Reproducible Shiny apps with shinymeta · ABACUS 2019 Nov 2019

Debug Large Data Apps: shiny + reactlog + sparklyr · Advanced R workshop, Northeastern University — May 2019
Reactlog 2.0: Debugging the State of Shiny · rstudio::conf(2019) · video — Jan 2019
Shiny · NASA Datanauts — Aug 2018
Cognostics: Metrics enabling detailed interactive visualization of big data · CSESC 2017 — Apr 2017
Web Scraping with R — Mar 2017
tidyverse[c("magrittr", "dplyr", "tidyr")] · Purdue Graduate Statistics Seminar — Jan 2017
Cognostics: Metrics enabling detailed interactive visualization of big data · JSM 2016 — Aug 2016
ggduo: Pairs plot for two group data · useR! 2016 — Jun 2016
Analysis and Visualization of Large Complex Data with Tessera · Spring Research Conference, IIT — Jun 2016
Analysis and Visualization of Large Complex Data with Tessera · Purdue Graduate Statistics Seminar — Apr 2016
Analysis and Visualization of Large Complex Data with Tessera · Iowa State Graphics Research Group — Mar 2016
Analysis and Visualization of Large Complex Data with Tessera · NUMBAT Seminar, Monash University — Mar 2016
gqlr = GraphQL + R: a data query language and runtime · WOMBAT 2016 — Feb 2016
Web Scraping with R · American Credit Acceptance — Oct 2015
Trelliscope: D&R visualization tool · Fields Institute Big Data Visualization · video — Feb 2015
ggplot2: displaying spatial and temporal data · Purdue Spatial Statistics Seminar — Oct 2014
git · Purdue Working Group — Jun 2014
Reducing Working Environment Inefficiencies · Purdue Graduate Statistics Seminar — Feb 2014
helpr: Help for R · Iowa State Working Group — Sep 2010
GGally: A Plot Matrix for All Variable Types · JSM 2010 — Aug 2010

Videos

Git Worktree-First Workflow + Conductor — Aug 2026
Shiny 1.2 Release: Wrap your data frame with Narwhals! — Oct 2024
Getting Started with {shinytest2} Part I · Example + basics — Jul 2022
Getting Started with {shinytest2} Part 3 · Using shiny.testmode in {shinytest2} — Jul 2022
Getting Started with {shinytest2} Part 2 · Exporting values — Jul 2022
Programming Games with Shiny · Roll the Dice: with Quosures! — Jun 2022
Programming Games with Shiny · Roll the Dice — May 2022
Programming Games with Shiny · Guess the Number — May 2022
Programming Games with Shiny · Dragon Realm — Mar 2022
Maximize computing resources using future_promise() — Feb 2022
{reactlog} Rundown — Feb 2022

Blog posts

Genomes, Prompts, and Shiny: What I Built During My Summer at Posit — Oct 2026
Introducing shinyreact: React UI backed by a Shiny server — Sep 2026
Shiny for Python 1.8 — Sep 2026
Shiny updates: R 1.14, Python 1.7, bslib 0.12 — Aug 2026
Bringing OpenTelemetry to R in production — May 2026
Shiny for Python 1.6 brings toolbars and OpenTelemetry — Apr 2026
OpenTelemetry + Shiny for R v1.12 — Dec 2025
Tool Calling UI in shinychat — Nov 2025
promises v1.5.0 — Nov 2025
Shiny at posit::conf(2025) — Sep 2025
Posit at SciPy 2025 — Jul 2025
Shiny 1.4 brings bookmarking and Generative AI docs — Apr 2025

Shiny for Python 1.2.0	Oct 2024
Testing Shiny Apps in Python	Oct 2024
Announcing Shiny for Python 1.0	Jul 2024
Editable data tables in Shiny for Python 0.9.0	May 2024
Introducing: Shiny Express	Jan 2024
Shiny for Python 0.6.1	Dec 2023
Shiny for Python 0.6.0	Nov 2023
Shiny for Python 0.5.0	Aug 2023
Shiny for Python 0.4.0	Jun 2023
Shiny for Python out of alpha	Apr 2023
plumber 1.1.0	Mar 2021
Shiny 1.6	Feb 2021
learnr 0.10.0	Dec 2019
leaflet 2.0.0	May 2018

Honors & awards

National Science Foundation Graduate Research Fellowship · the only Computational Statistics awardee of 13 statistics recipients in 2012 — 2012 – 2017

Visiting Scientific Researcher, Monash University · large data exploration tools with NUMBAT — 2016

Google Summer of Code, GGally::ggduo · pairs plots for two groups of variables — 2016

Best Dynamic / Interactive Visualization, Data Insight SF — 2011

NSF Graduate Research Fellowship, honorable mention — 2011

ASA Data Expo, 2nd place (Iowa State working group) — 2009

Teaching & mentoring

Internship Supervisor, Posit · Samuel Bharti: a bioinformatics Shiny showcase, 10 Shiny apps built on 5 packages — 2026

Workshop Teaching Assistant, rstudio::conf / posit::conf · Shiny in Production (2019), Applied Machine Learning (2020), Level Up with Shiny for R (2024), Shiny for R (2025) — 2019 – 2025

Internship Advisor, RStudio · Nischal Shrestha: gradethis, a companion package for grading learnr exercises — 2020

Internship Advisor, RStudio · JooYoung Seo: accessibility within Shiny for R — 2020

Internship Supervisor, RStudio · Leslie Huang: leaflet.providers and education materials — 2019

Internship Advisor, RStudio · Daniel Chen: gradethis, a companion package for grading learnr exercises — 2019

Instructor, R for Data Science Seminar, Purdue University — 2017

Teaching Assistant, Big Data Analytics: Statistics and Data Visualization, Purdue University — 2017

Teaching Assistant, Hadley Wickham's R Master Class, San Francisco and Chicago — 2011 – 2015

Service

President, Statistics Graduate Student Office, Purdue University · Ran a town hall, a graduate student mentor program, and an esteemed speaker event with Arthur Dempster. — 2016 – 2017

Graduate Student Mentor, Purdue University Statistics — 2014 – 2016

StatCom Assistant, Purdue University — 2014 – 2016

Skills

Languages	R, Python, TypeScript, JavaScript, HTML, CSS, SQL, Bash
Frameworks	Shiny (R and Python), React, plumber, htmltools, ggplot2, tidyverse
Testing & CI	Playwright, pytest, testthat, shinytest2, GitHub Actions
Publishing	CRAN, PyPI, npm, conda-forge, pkgdown, Quarto, GitHub Pages